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Final Exam
International Campus, MD Program
Respiratory System
February 2022 semester

Name and Surname:

Student ID Number:

Course Coordinator: Dr. Pasbakhsh

Date of Exam: Jul 04, 2022

Duration of Exam: 45 Minutes

Key

MCQ Question	34
Short Answer Questions	6

Directions:

Each of the following questions contains one correct answer only. •

Please choose the correct answer for the MCQs in the answer sheet with pencil. •

For short answer Questions, write down your answer in the question pamp •

physiology

Which muscle contraction makes the pleural pressure more positive?

1

Internal intercostal

Sternocleidomastoid

الف

External intercostal

ب

Diaphragm

ج

د

متنوع:

physiology

According to the V/Q mismatch, what is the difference between the oxygen pressure of the lower, middle and upper alveoli in a healthy and upright subject?

2

Oxygen pressure in the lower alveoli is higher than the middle and upper

الف

Oxygen pressure in the upper alveoli is higher than the middle and lower

ب

Oxygen pressure in the middle alveoli is higher than the lower and upper

ج

Oxygen pressure is the same in all alveoli

د

متنوع:

physiology

What is the effect of pleural pressure changes on alveolar pressure during inspiration?

3

The more negative the pleural pressure, the more positive the alveolar pressure

الف

The more positive the pleural pressure, the more positive the alveolar pressure

ب

The more positive the pleural pressure, the more negative the alveolar pressure

ج

The more negative the pleural pressure, the more negative the alveolar pressure

د

متنوع:

12

physiology

Which of the following IS NOT correct about anatomical shunt?		9
Oxygen pressure decreases from 104 to 95 mm Hg		
Arterial blood CO ₂ pressure increases from 40 to 45 mm Hg	الف	
Pulmonary vein and bronchial vein blood are mixed	ب	
It forms part of the physiological shunt	ج	
	د	
	منبع:	

physiology

Inhibition of beta-receptors in the airways leads to an increase in which of the following?		10
Anatomical dead space		
Surface tension	الف	
Pleural pressure	ب	
Airway resistance	ج	
	د	
	منبع:	

physiology

What is the role of Haldan's effect in the respiratory system?		11
It increases the release of O ₂ from hemoglobin	الف	
It increases the excretion of CO ₂ from the lungs	ب	
It reduces the tendency of Fe ²⁺ + hemoglobin for binding with oxygen	ج	
It reduces the diffusion of CO ₂ from peripheral tissue into the capillaries	د	
	منبع:	

physiology		
What part of the lung in upright subject is zone 2 blood flow during exercise?		6
Top of the lung	الف	
Middle of the lung	ب	
Bottom of the lung	ج	
Zone 2 does not exist	د	
		متبع:

physiology		
Diffusion coefficient of a gase in a liquid depends on which following factor?		7
Pressure difference between the two ends of the diffusion path	الف	
Diffusion distance	ب	
Gas solubility	ج	
Cross-sectional area	د	
		متبع:

physiology		
If the thickness of the respiratory membrane doubles due to pulmonary fibrosis, what will be CO2 diffusing capacity, approximately?		8
50 ml/min/mmHg	الف	
100 ml/min/mmHg	ب	
200 ml/min/mmHg	ج	
300 ml/min/mmHg	د	
		متبع:

physiology

If in a premature infant, the size of the alveoli is one-third of the normal value and the amount of surface tension is tripled due to the low production of surfactant, how does the collapse pressure change compared to the normal infant?		4
It triples		
It increases six times	الف	
It increases nine times	ب	
It does not change	ج	
	د	
		متنوع

physiology

If lung compliance and thoracic compliance are the same and assumed to be 300 ml / CmH ₂ O, what would be the total pulmonary system compliance as ml / CmH ₂ O?		5
150		
300	الف	
450	ب	
600	ج	
	د	
		متنوع

physiology		
Which respiratory center provides strong signals for contraction of the rectus abdominis muscle to increase pulmonary ventilation in deep expiration?		12
VRG	الف	
DRG	ب	
Pneumotoxic center	ج	
Chemosensitive area in medulla	د	
		منبع:

physiology		
What is the initial pressure difference for O ₂ diffusion from alveoli into the pulmonary capillaries?		13
64 mmHg	الف	
21 mmHg	ب	
11 mmHg	ج	
5 mmHg	د	
		منبع:

physiology		
What is the effect of using acetazolamide as an inhibitor of carbonic anhydrase on respiratory gases?		14
A decrease in arterial blood O ₂ pressure	الف	
An increase in the O ₂ pressure in interstitial fluid	ب	
A decrease in arterial blood CO ₂ pressure	ج	
An increase in the CO ₂ pressure in interstitial fluid	د	
		منبع:

physiology	
What is the role of the pneumotoxic center in regulation of respiration?	15
Stimulation of DRG	
Stimulation of VRG	الف
Inhibition of DRG	ب
Inhibition of VRG	ج
	د
منبع:	

physiology	
Which factor DOSE NOT stimulate the chemosensitive area in the medulla?	16
Increased H ⁺ concentration in the cerebrospinal fluid	الف
Decreased O ₂ concentration in blood	ب
Increased CO ₂ concentration in interstitial fluid of medulla	ج
Increased CO ₂ concentration in the blood	د
منبع:	

	Histology	Question time (seconds): →
17	In Structural Histology Which Part Of Nasal cavity Is Observed Goblet Cell?	
	1	Vestibuli
	2	Choncea
	3	Palate
	4	Respiratory
Source:		
Description:		

Histology		Question time (seconds): →
18	Which Part Of The Air Ducts Is Observed True Elastic Fiber?	
1	Pharynx Tissue	
2	Broncos	
3	Thracea	
4	Larynex	

Source:	
Description:	

	Histology	Question time (seconds): →
19	What Is The Name Macrophage In Lung Tissue?	
	1 Dust Cell	
	2 Goblet Cell	
	3 Columnar Cell	
	4 celara Cell	
Source:		
Description:		

	Histology	Question time (seconds): →
20	Which Part Of The Lung Is Observed Celera Cell?	
	1 Thracea	
	2 Broncos	
	3 Alveolar Ducts	
	4 Bronchiole	
Source:		
Description:		

	Histology	Question time (seconds): →
21	In Structural Histology Which Of Alveolar Cell Has a Circular Shaped Nuclease And granular in the Cytoplasm?	
	1 Pneumocyte II	
	2 Pneumocyte I	
	3 Pneumocyte IV	
	4 Pneumocyte III	
Source:		
Description:		

Anatomy

Nasal septum consists of all of the following items except		22
Septal nasal cartilage		
Vomer bone	الف	
Perpendicular plate of Ethmoid bone	ب	
Lacrimal bone	ج	
	د	
		منبع:

Anatomy

All of the following statement about Maxillary sinus are correct except.....		23
Roof of the sinus forms floor of the orbit	الف	
Maxillary sinus opening opens in middle meatus	ب	
It is the largest paranasal sinus	ج	
The floor of the Maxillary sinus related to the Sphenopalatin foramen	د	
		منبع:

Anatomy

Posterior inferior nasal nerve is branch of which of the following nerve?		24
Nasopalatin nerve	الف	
Greater Palatin nerve	ب	
Anterior Ethmoidal nerve	ج	
Posterior Ethmoidal nerve	د	
		منبع:

Anatomy		
All of the following cartilage has articular facet except....?		25
Thyroid	الف	
Cricoid	ب	
Epiglottis	ج	
Arythnoid	د	
		منبع:

Anatomy		
All of the following statement about larynx is correct except..		26
Middle cavity of larynx limited above by vestibular folds	الف	
Middle cavity of larynx expanded lateraly to ventricle	ب	
Laryngeal saccule open to vestibule	ج	
Mucus glands in saccules lubricates vocal folds	د	
		منبع:

Anatomy		
All of the following items attached to arythnoid cartilage except.		27
Hyoid bone	الف	
Vocal ligaments	ب	
Quadrangular membrane	ج	
Vestibular ligaments	د	
		منبع:

Anatomy		
All of the following arteries supply nasal cavity except..		28
Superior labial		
Greater palatin	الف	
Internal Carotid	ب	
Superior thyroid	ج	
	د	
		منبع:

Anatomy		
All of the following nerves innervate nasal cavity except..		29
Olfactory	الف	
Ophthalmic	ب	
Recurrent laryngeal	ج	
Maxillary	د	

Anatomy		
Anterior margin of Laryngeal inlet forms by		30
Epiglottis	الف	
Aryepiglottic folds	ب	
Vestibular ligament	ج	
Interarytenoid notch	د	

Anatomy		
Left lung has relation to all of the following structures except...		31
Spleen	الف	
Arch of the Azygous vein	ب	
Oesophagus	ج	
Thoracic Aorta	د	

Anatomy		
In surface Anatomy in mid axillary line shadow of plura is at level of which rib		32
4	الف	
6	ب	
8	ج	
10	د	

Embryology		
The cartilagnious, muscular, and connective tissue components of the trachea and lungs are derived from ----- surrounding the foregut		33
Splanchnic mesoderm	الف	
Endoderm	ب	
Neural crest cells	ج	
2 th pharyngeal arch	د	
منبع:		

Embryology		
By the end of the 6 month approximately how many generations of the subdivisions have formed in the lungs		34
12	الف	
6	ب	
17	ج	
4	د	
منبع:		